AMRUTVAHINI COLLEGE OF ENGINEERING, SANGAMNER

(An Autonomous Institute)

S.Y. B. Tech (Electronics and Computer Engineering)

CCE III– Course Project, Semester – III, Oct. 2026

Data Structures and Algorithms [R25-EC-PCC-202]

Unit V: Trees and Graphs (Total: 08 Marks)

Activity- Develop a C++-based application/project that solves a real-world problem using Trees and/or Graphs. Students may use AI tools for assistance but must understand, test, modify and explain the generated code.

General Guidelines

1. Team Size:
A group of up to 03 students.

2. Selection of Topic:
Students must develop a C++-based interactive application/project that solves a real-world problem using Trees and/or Graphs. The use of Trees and/or Graphs must be relevant and justified in the selected project.

The project must be interactive (menu-based).

Suggested Domains:

- File and Directory Management using Trees
- Organization Hierarchy Management using Trees
- Shortest Path Finder using Graphs/Road Network and Route Finding using Graphs
- Social Network Connectivity Analysis using Graphs
- Family Tree Management

3. Mandatory Technical Requirements:

- Implement at least one of the following data structures: Tree or Graph**.**
- Apply suitable operations or algorithms related to the selected data structure, such as tree traversal, insertion, deletion, searching, graph traversal (BFS/DFS), or shortest path finding, wherever applicable.
- Use appropriate searching and/or sorting algorithms, wherever relevant to the project.
- Include basic object-oriented programming concepts such as classes and constructors. Operator overloading may be included wherever appropriate.
- Develop the application in C++ with proper code organization, meaningful variable names and comments.
- Students may use AI tools for assistance but must understand, test, modify and explain the generated code.

4. Documentation Must Include:

- Project Title and Abstract
- Problem Statement and Objectives
- Data Structures Used and Justification
- System Design / Flowchart / Class Diagram
- Algorithms Used and Their Explanation
- Source Code with Comments
- Sample Input and Output
- Conclusion and Future Scope

Note - Source Code with Comments, Sample Input and Output as a print, rest of the things should be handwritten. Single report is to be submitted by the group.

5. Demonstration and Evaluation:
Students must demonstrate the working application, explain the implementation of Trees and/or Graphs, and answer questions related to the data structures and algorithms used in the project.

Note - STL Restriction: The use of C++ Standard Template Library (STL) containers and data structures, such as vector, list, stack, queue, map, and set, is not permitted. Students must implement the required Trees and/or Graphs and their operations manually.